

Bessel's Equation & Bessel's Function

The differential equation, $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2)y = 0$,

is called the Bessel's differential equation and particular solution of this differential equation are called Bessel's functions of order 'n'.

Solution of Bessel's Differential Equation:

$$x^2 y'' + x y' + (x^2 - n^2)y = 0 \quad \text{--- (1)}$$

$x=0$ is a regular singular point.

$$\text{Let, } y = \sum_{k=0}^{\infty} a_k x^{m+k} \quad \text{--- (2)}$$

$$y' = \sum_{k=0}^{\infty} a_k (m+k) x^{m+k-1}$$

$$y'' = \sum_{k=0}^{\infty} a_k (m+k)(m+k-1) x^{m+k-2}$$

Putting these values in eqⁿ. (1),

$$\begin{aligned} & \sum a_k (m+k)(m+k-1) x^{m+k-2} \cdot x^2 \\ & + \sum a_k (m+k) x^{m+k-1} \cdot x \\ & + \sum a_k x^{m+k} (x^2 - n^2) = 0 \end{aligned}$$

$$\Rightarrow \sum a_k(m+k)(m+k-1)x^{m+k} + \sum a_k(m+k)x^{m+k} + \sum a_k x^{m+k+2} - \sum a_k n^2 x^{m+k} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} \{(m+k)^2 - n^2\} a_k x^{m+k} + \sum_{k=0}^{\infty} a_k x^{m+k+2} = 0$$

The lowest power of x is x^m when k is zero....

Equating to zero the coefficient of x^m , we get,

$$m^2 - n^2 = 0 \quad \text{as } a_0 \neq 0 \quad [k=0]$$

$$\Rightarrow m = \pm n$$

Equating to zero the coefficient of next term i.e. x^{m+1} , we get,

$$[(m+1)^2 - n^2] a_1 = 0 \quad [k=1]$$

$$\Rightarrow a_1 = 0$$

Equating to zero the coefficient of x^{m+k+2} ,

$$a_{k+2} = \frac{-a_k}{(m+k+2)^2 - n^2} \quad (3) \quad [k=k+2]$$

$$[\{(m+k+2)^2 - n^2\} a_{k+2} + a_k] x^{m+k+2} = 0$$

Putting $k=1$ in eqⁿ. (3),

$$a_{1+2} = a_3 = \frac{-a_1}{(m+3)^2 - n^2} = 0$$

Similarly, $a_5 = a_7 = \dots = 0$ for $k=1, 3, 5, \dots$

Putting $k=0$ in eqⁿ. (3),

$$a_{0+2} = a_2 = \frac{-a_0}{(m+2)^2 - n^2}$$

$$\text{Similarly, } a_4 = \frac{-a_2}{(m+4)^2 - n^2} \quad [k=2]$$

$$\text{i.e. } a_4 = \frac{a_0}{[(m+2)^2 - n^2][(m+4)^2 - n^2]}$$

So, for $k=0, 2, 4, \dots$

$$a_2 = \frac{-a_0}{(m+2)^2 - n^2}, \quad a_4 = \frac{a_0}{[(m+2)^2 - n^2][(m+4)^2 - n^2]}, \dots$$